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A Unified Framework for Rerandomization using Quadratic Forms

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ABSTRACT

When designing a randomized experiment, one way to ensure treatment and control groups exhibit similar covariate distributions is to randomize treatment until some prespecified level of covariate balance is satisfied; this strategy is known as rerandomization. Most rerandomization methods use balance metrics based on a quadratic form $\mathbf{v}^T \mathbf{A} \mathbf{v}$, where $\mathbf{v}$ is a vector of covariate mean differences and $\mathbf{A}$ is a positive semi-definite matrix. In this work, we derive general results for treatment-versus-control rerandomization schemes that employ quadratic forms for covariate balance. In addition to allowing researchers to quickly derive properties of rerandomization schemes not previously considered, our theoretical results provide guidance on how to choose $\mathbf{A}$ in practice. We find the Mahalanobis and Euclidean distances optimize different measures of covariate balance. Furthermore, we establish how the covariates' eigenstructure and their relationship to the outcomes dictate which matrix $\mathbf{A}$ yields the most precise difference-in-means estimator for the average treatment effect. We find the Euclidean distance is minimax optimal, in the sense that the difference-in-means estimator's precision is never too far from the optimal choice. We verify our theoretical results via simulation and a real data application, and demonstrate how the choice of $\mathbf{A}$ impacts the variance reduction of rerandomized experiments. Supplementary materials for this article are available online, including a standardized description of the materials available for reproducing the work.

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1. Introduction

In the design stage of randomized experiments, it is important to address covariate imbalance between treatment and control groups. Large covariate imbalances can lead to increased standard errors when estimating causal effects—this can reduce statistical power and make it more difficult to interpret results (Lachin 1988; Senn 1989; Lin, Zhu, and Su 2015; Branson, Li, and Ding 2024). Therefore, it is often preferable to ensure treatment groups exhibit covariate balance before the experiment is conducted. One classical experimental design strategy for ensuring covariate balance is blocking, where treatment is randomized within groups of subjects with similar categorical covariates (Box, Hunter, and Hunter 1978; Pashley and Miratrix 2021). However, blocking cannot be easily extended to accommodate many continuous variables. Instead, an experimental design tool that can handle categorical or continuous variables is rerandomization, where subjects are randomized until some prespecified level of covariate balance is achieved. Although the concept of rerandomization had been discussed as early as Fisher (Fisher 1992), it was not until Morgan and Rubin (2012) that a theoretical framework was established in which covariate balance is measured using the Mahalanobis distance between the covariate means of the treatment and control groups (often abbreviated as “ReM”; hereafter, we simply use “Mahalanobis Rerandomization”). Since then, there have been many extensions that also use the Mahalanobis distance, including those for

tiers of covariates that vary in importance (Morgan and Rubin 2015), factorial designs (Branson, Dasgupta, and Rubin 2016; Li, Ding, and Rubin 2020), sequential designs (Zhou et al. 2018), clustered experiments (Lu et al. 2023), and Bayesian designs (Liu et al. 2025).

One key property of Mahalanobis Rerandomization is that it reduces the variance of all covariate mean differences by an equal amount. While this can be advantageous in some contexts, placing equal weight on all covariates can lead to poor performance when covariates are high-dimensional. One option is to define tiers of covariates based on variable importance, as suggested in Morgan and Rubin (2015). However, it is necessary to specify which covariates are most important, which can be difficult to do in practice. Two methods that automatically place importance on particular covariates in a high-dimensional space are Ridge Rerandomization (Branson and Shao 2021) and PCA Rerandomization (Zhang, Yin, and Rubin 2024). Ridge Rerandomization uses a ridge penalty within the Mahalanobis distance to diagonalize the inverse covariance matrix, thereby placing more importance on top eigenvectors. Meanwhile, PCA Rerandomization uses the Mahalanobis distance only on the top k principal components. This applies a greater amount of variance reduction to the top k components than classical Mahalanobis Rerandomization, but no reduction to the bottom $d - k$ components, where d is the dimension of the covariates.

Most rerandomization methods—including Ridge Rerandomization and PCA Rerandomization—use some kind of

quadratic form $\mathbf{v}^T \mathbf{A} \mathbf{v}$, where $\mathbf{v} \in \mathbb{R}^d$ is a vector of covariate mean differences and $\mathbf{A} \in \mathbb{R}^{d \times d}$ is a positive semi-definite matrix. For example, if we let Σ denote the covariance matrix of the covariate mean differences, $\mathbf{A} = \Sigma^{-1}$ corresponds to Mahalanobis Rerandomization and $\mathbf{A} = (\Sigma + \lambda \mathbf{I}_d)^{-1}$ corresponds to Ridge Rerandomization. To our knowledge, this general framing of quadratic forms for rerandomization has only been noted in Lu et al. (2023) and remains largely unexplored. To fill this gap, we make several contributions. First, we derive general results for any treatment-versus-control rerandomization scheme that uses a quadratic form as its balance metric, as compared to a completely randomized experiment. This allows us to more quickly rederive previous results in the literature, as well as derive results for rerandomization schemes that have not been previously considered. Second, we establish guidance on how to optimally choose the matrix $\mathbf{A}$ in practice. While others such as Liu et al. (2025) and Lu et al. (2023) have considered optimal rerandomization schemes for minimizing the variance of the difference-in-means estimator for the average treatment effect, their results require knowledge about how covariates are related to outcomes in order to be implemented in practice, which is often not available before the start of the experiment. Thus, one of our contributions is deriving optimal rerandomization schemes when outcome information is not available. We show that the Mahalanobis distance maximizes the total variance reduction across covariates and the Euclidean distance minimizes the Frobenius norm of the covariate mean differences' covariance matrix. Furthermore, we establish how the covariates' eigenstructure and their relationship to the outcomes dictate which matrix $\mathbf{A}$ yields the most precise difference-in-means estimator for the average treatment effect. We find that the Euclidean distance is minimax optimal, in the sense that the difference-in-means estimator's precision is never too far from the optimal choice, regardless of the relationship between covariates and outcomes. Consequently, our results are useful for practitioners who seek guidance on which rerandomization method is most appropriate for their dataset.

That said, there are limitations to our work. First, not every rerandomization method can be expressed using quadratic forms. For example, Zhao and Ding (2024) consider randomizing treatment until the p -values for statistical tests of covariate imbalance are above some threshold; while some of these tests can be written as a quadratic form (e.g., their joint acceptance rules), other tests cannot, making it unclear how they compare to the rerandomization procedures we consider in this work. Second, there are many experimental design strategies that cannot be framed as a rerandomization procedure; Kallus (2018) suggest covariate balancing through kernel allocation, Li, Kang, and Huang (2021) suggest partitioning experimental units based on kernel density estimates, and Harshaw et al. (2024) suggest optimizing a tradeoff between robustness and covariate balance. Although we establish optimality results for rerandomization schemes involving quadratic forms, our results do not suggest whether alternative experimental design strategies may be preferable. Nonetheless, our work considers a broad class of rerandomization schemes, and furthermore provides guidance on how researchers can choose designs within this class in practice.

The remainder of the article is as follows. In Section 2 we define all important notation used in the article. In Section 3 we review existing rerandomization methods, with a focus on methods whose balance metrics can be expressed as a quadratic form. In Section 4 we derive general results for the covariance of the covariate mean differences after rerandomization using quadratic forms, including optimality results for covariate balance and variance reduction for the difference-in-means estimator. In Section 5 we validate our theoretical results via simulation and a real data application; we find that Euclidean Rerandomization's performance is robust across many different settings. In Section 6 we conclude with a discussion of our results and directions for future work.

2. Notation

Let $\mathbf{X} = (\mathbf{X}_1, \dots, \mathbf{X}_n)^T \in \mathbb{R}^{n \times d}$ be the covariate matrix representing n experimental units, where $\mathbf{X}_i \in \mathbb{R}^d$ denotes the vector of d covariates for subject i . To make theoretical results notationally succinct, we assume that the columns of $\mathbf{X}$ have been centered and standardized such that each column has mean zero and variance one. Next, define $\mathbf{W} = (W_1, \dots, W_n)^T \in \mathbb{R}^n$ to be the treatment assignment vector where $W_i = 1$ if unit i has been assigned treatment and $W_i = 0$ otherwise. Let $n_1 = \sum_{i=1}^n W_i$ and $n_0 = \sum_{i=1}^n (1 - W_i)$ be the number of units in treatment and control, respectively, and $p = \frac{1}{n} \sum_{i=1}^n W_i$ be the proportion of treated subjects. We use the potential outcomes framework, where unit i has fixed potential outcomes, $Y_i(1)$ and $Y_i(0)$, denoting their outcomes when $W_i = 1$ and $W_i = 0$. Thus, we assume the stable unit treatment value assumption holds, such that one subject's outcome does not depend on another subject's treatment assignment, that is, no interference. Our goal is to estimate the average treatment effect, given by $\tau = \frac{1}{n} \sum_{i=1}^n (Y_i(1) - Y_i(0))$ using the difference-in-means estimator, $\hat{\tau} = \frac{1}{n_1} \sum_{i=1}^n W_i Y_i(1) - \frac{1}{n_0} \sum_{i=1}^n (1 - W_i) Y_i(0)$. To measure covariate balance, we consider the covariate mean differences, defined as $\hat{\tau}_{\mathbf{X}} := \bar{\mathbf{X}}_T - \bar{\mathbf{X}}_C = \frac{1}{n_1} (\mathbf{X}^T \mathbf{W}) - \frac{1}{n_0} (\mathbf{X}^T (\mathbf{1}_n - \mathbf{W}))$ where $\mathbf{1}_n \in \mathbb{R}^n$ is a vector whose coefficients are all equal to one. We consider the covariates $\mathbf{X}$, the proportion of treated subjects p , and potential outcomes as fixed, such that the only stochastic element is the treatment assignment $\mathbf{W}$. Next, let Σ denote the covariance matrix of the covariate mean differences when $\mathbf{W}$ is assigned according to complete randomization, where a random n_1 units are assigned to treatment and n_0 units are assigned to control. As shown in Morgan and Rubin (2012), Σ is a fixed matrix that can be expressed in terms of the sample covariance of $(\mathbf{X}_1, \dots, \mathbf{X}_n)$,

$$\begin{aligned} \Sigma &= \text{cov}(\sqrt{n}(\bar{\mathbf{X}}_T - \bar{\mathbf{X}}_C) \mid \mathbf{X}) \\ &= \frac{\frac{1}{n-1} \sum_{i=1}^n (\mathbf{X}_i - \bar{\mathbf{X}})(\mathbf{X}_i - \bar{\mathbf{X}})^T}{p(1-p)}. \end{aligned}$$

Following other works (e.g., Li, Ding, and Rubin 2018), we use the scaling term $\sqrt{n}$ in the definition of Σ to simplify asymptotic results in Section 4. Furthermore, throughout we condition on $\mathbf{X}$ to emphasize that the only random variable is treatment assignment. Because the potential outcomes are fixed, we implicitly condition on them as well. We define $\mathbf{S}_+^d = \{\mathbf{M} \in \mathbb{R}^{d \times d} \mid$

$\mathbf{M} \succeq 0, \mathbf{M} = \mathbf{M}^T$ to be the set of symmetric, positive semi-definite matrices. We use $\lambda_1, \dots, \lambda_d$ to denote the eigenvalues of Σ and Γ to denote the orthogonal matrix of eigenvectors of Σ . Similarly, we use $\eta_1, \dots, \eta_d$ for the eigenvalues of $\Sigma^{1/2} \mathbf{A} \Sigma^{1/2}$ and Ω for its eigenvectors. We use $\text{diag}\{(a_j)_{1 \leq j \leq d}\}$ to refer to a diagonal matrix with elements $a_1, \dots, a_d$. Finally, for $\mathbf{x} \in \mathbb{R}^d$ we define $\|\mathbf{x}\|_2^2 = \sum_{j=1}^d x_j^2$ to be the squared L_2 norm.

3. Review of Rerandomization Methods

Here, we review existing rerandomization methods that use particular quadratic forms to balance treatment and control groups. While there are other rerandomization methods that do not rely on quadratic forms such as the marginal acceptance rules defined in Zhao and Ding (2024), we focus less on these as our goal is to establish results for distance metrics that can be written as quadratic forms. Along the way, we note how each method incorporates the principal components of $\mathbf{X}$ and the eigenstructure of Σ , that is, the covariance of $\sqrt{n}\hat{\tau}_{\mathbf{X}}$ under complete randomization. As we show in Section 4, the eigenstructure plays a crucial role in determining which rerandomization method is optimal for reducing the variance of the difference-in-means estimator $\hat{\tau}$.

3.1. Mahalanobis Rerandomization

First investigated by Morgan and Rubin (2012) and extended to many other experimental design settings (Morgan and Rubin 2015; Branson, Dasgupta, and Rubin 2016; Zhou et al. 2018; Li, Ding, and Rubin 2020; Li and Ding 2020; Wang and Li 2022; Lu et al. 2023; Wang, Wang, and Liu 2023; Shi, Zhao, and Liu 2025), Mahalanobis Rerandomization balances treatment and control groups by randomizing until $M \leq a$, where a is a prespecified threshold and $M = (\sqrt{n}\hat{\tau}_{\mathbf{X}})^T \Sigma^{-1} (\sqrt{n}\hat{\tau}_{\mathbf{X}})$ is the Mahalanobis distance between $\bar{\mathbf{X}}_T$ and $\bar{\mathbf{X}}_C$. Morgan and Rubin (2012) establish several key properties of Mahalanobis Rerandomization that serve as benchmarks for rerandomization research. First, they show that the difference-in-means estimator $\hat{\tau}$ remains unbiased in finite samples conditional on $M \leq a$ provided that $\sum_{i=1}^n W_i = \sum_{i=1}^n (1 - W_i)$. Furthermore, the expectation of all observed and unobserved covariate mean differences is still zero under rerandomization. Next, the authors show that Mahalanobis Rerandomization applies an equal-percentage variance reduction to all covariates, in the sense that $\text{cov}(\sqrt{n}\hat{\tau}_{\mathbf{X}} | \mathbf{X}, M \leq a) = v_a \text{cov}(\sqrt{n}\hat{\tau}_{\mathbf{X}} | \mathbf{X})$ where $v_a = \mathbb{P}(\chi_{d+2}^2 \leq a) / \mathbb{P}(\chi_d^2 \leq a) \leq 1$ is the variance reduction term, that is, the amount that rerandomization reduces the variances and covariances of $\hat{\tau}_{\mathbf{X}}$, compared to complete randomization. Finally, when the treatment effect is additive, Mahalanobis Rerandomization reduces the variance of $\hat{\tau}$ by $1 - (1 - v_a)R^2$ in finite samples, where R^2 is the squared multiple correlation between the potential outcomes and $\mathbf{X}$, that is, $R^2 = \text{cov}(\hat{\tau}, \hat{\tau}_{\mathbf{X}} | \mathbf{X}) \text{cov}(\hat{\tau}_{\mathbf{X}} | \mathbf{X})^{-1} \text{cov}(\hat{\tau}_{\mathbf{X}}, \hat{\tau} | \mathbf{X}) / \text{var}(\hat{\tau} | \mathbf{X})$. Li, Ding, and Rubin (2018) show that these results hold asymptotically under non-additivity and unequal sample sizes.

Despite these useful properties, placing equal priority on all covariates creates challenges in high-dimensional settings because the variance reduction term v_a is increasing in d ; as

$d \rightarrow \infty$ then $v_a \rightarrow 1$, such that there is no variance reduction. Consequently, others have recommended balancing metrics that place higher priority on smaller-dimensional spaces.

3.2. Ridge Rerandomization

To address the problems that Mahalanobis Rerandomization suffers from in high-dimensional settings, Branson and Shao (2021) introduce a ridge penalty $\lambda \geq 0$ to the Mahalanobis distance. They suggest randomizing until

$$M_\lambda = (\sqrt{n}\hat{\tau}_{\mathbf{X}})^T (\Sigma + \lambda \mathbf{I}_d)^{-1} (\sqrt{n}\hat{\tau}_{\mathbf{X}}) \quad (1)$$

is less than some prespecified $a_\lambda > 0$. Under Ridge Rerandomization, there is no longer an equal-percentage variance reduction to the covariance of the covariate mean differences. Instead, $\text{cov}(\sqrt{n}\hat{\tau}_{\mathbf{X}} | \mathbf{X}, M_\lambda \leq a_\lambda) = \Gamma \Lambda^{1/2} (\text{diag}\{(d_{j,\lambda})_{1 \leq j \leq d}\}) \Lambda^{1/2} \Gamma^T$ where Γ is the matrix of eigenvectors of Σ , Λ is the diagonal matrix of eigenvalues of Σ , and

$$d_{j,\lambda} = \mathbb{E} \left[\mathcal{Z}_j^2 | \mathbf{X}, \sum_{\ell=1}^d \frac{\lambda_\ell}{\lambda_\ell + \lambda} \mathcal{Z}_\ell^2 \leq a_\lambda \right] \quad (2)$$

where $\mathcal{Z}_1, \dots, \mathcal{Z}_d \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$ and $d_{1,\lambda} \leq \dots \leq d_{d,\lambda} \leq 1$ are variance reduction terms. Importantly, because the covariance matrix under Mahalanobis Rerandomization can be written as $\text{cov}(\sqrt{n}\hat{\tau}_{\mathbf{X}} | \mathbf{X}, M \leq a) = \Gamma \Lambda^{1/2} (v_a \mathbf{I}_d) \Lambda^{1/2} \Gamma^T$ we can see that these two methods differ by the variance reduction they apply to the eigenvectors of Σ (or equivalently, the principal components of $\mathbf{X}$). Ridge Rerandomization applies a greater variance reduction to the top eigenvectors of Σ , but a lesser variance reduction to the bottom eigenvectors as weighted by $\lambda_j / (\lambda_j + \lambda)$ where $\lambda_1 \geq \dots \geq \lambda_d$. This differs from Mahalanobis Rerandomization, which reduces the variance of each eigenvector equally. As a consequence, the variance of $\hat{\tau}$ tends to be less under Ridge Rerandomization than under Mahalanobis Rerandomization in high-dimensional settings. However, Ridge Rerandomization does not strictly dominate Mahalanobis Rerandomization because, ultimately, the precision of $\hat{\tau}$ after rerandomization depends on the relationship between the principal components and the potential outcomes. Furthermore, Ridge Rerandomization requires selecting the tuning parameter λ , which can be computationally intensive.

3.3. PCA Rerandomization

PCA Rerandomization, introduced by Zhang, Yin, and Rubin (2024), also attempts to improve upon Mahalanobis Rerandomization in high-dimensional settings. Instead of adding a penalization term, PCA Rerandomization only considers the top k principal components. Let $\mathbf{X} = \mathbf{U} \mathbf{D} \mathbf{V}^T$ be the singular value decomposition of $\mathbf{X}$, where $\mathbf{U} \in \mathbb{R}^{n \times d}$ and $\mathbf{V} \in \mathbb{R}^{d \times d}$ are the orthogonal matrices of left and right singular vectors, and $\mathbf{D}$ is a diagonal matrix of singular values. Then, $\mathbf{Z} = \mathbf{U} \mathbf{D}$ is the matrix of principal components of $\mathbf{X}$ and $\mathbf{Z}_k = \mathbf{U}_k \mathbf{D}_k = (\mathbf{Z}_1^{(k)}, \dots, \mathbf{Z}_n^{(k)})^T \in \mathbb{R}^{n \times k}$ is the matrix of the top k principal components of $\mathbf{X}$. Then, PCA Rerandomization randomizes until

$$M_k = \{ \sqrt{n}(\bar{\mathbf{Z}}_T^{(k)} - \bar{\mathbf{Z}}_C^{(k)}) \}^T \Sigma_Z^{-1} \{ \sqrt{n}(\bar{\mathbf{Z}}_T^{(k)} - \bar{\mathbf{Z}}_C^{(k)}) \} \quad (3)$$

is less than some prespecified $a_k > 0$, where $\Sigma_Z = \text{cov}(\mathbf{Z}_k)/p(1-p)$ is the covariance matrix of $\sqrt{n}(\bar{\mathbf{Z}}_T^{(k)} - \bar{\mathbf{Z}}_C^{(k)})$. The authors show that

$$\begin{aligned} \text{cov}(\sqrt{n}\hat{\boldsymbol{\tau}}_X \mid \mathbf{X}, M_k \leq a_k) &= C_n \mathbf{V} \begin{pmatrix} v_{a_k} \mathbf{D}_k^2 & 0 \\ 0 & \mathbf{D}_{d-k}^2 \end{pmatrix} \mathbf{V}^T \\ &= \Gamma \Lambda^{1/2} \begin{pmatrix} v_{a_k} \mathbf{I}_k & 0 \\ 0 & \mathbf{I}_{d-k} \end{pmatrix} \Lambda^{1/2} \Gamma^T \end{aligned}$$

where $v_{a_k} = \mathbb{P}(\chi_{k+2}^2 \leq a_k)/\mathbb{P}(\chi_k^2 \leq a_k)$ and $C_n = [(n-1)p(1-p)]^{-1}$. Thus, PCA Rerandomization is still an equal-percentage variance reduction method, but only for the top k principal components. This introduces a tradeoff: for $k < d$ it follows that $v_{a_k} < v_{a_d}$, so there is a greater variance reduction to the top k principal components than if the full set of covariates were included during rerandomization. However, there is a loss of reduction to the bottom $d-k$ components. Again, depending on the relationship between principal components and potential outcomes, different rerandomization schemes will be preferable.

Although the variance of $\hat{\boldsymbol{\tau}}$ tends to be less under PCA Rerandomization than under Mahalanobis Rerandomization in high-dimensional settings, it also requires choosing the number of principal components to include. The authors suggest either choosing a fixed amount of variation explained by the top k principal components (e.g., 50%, 70%, or 90%) or using the Kaiser rule, which takes the top components whose variation is larger than the average amount of variance explained. In Section 4.4 we introduce new decision rules for the number of principal components to keep based on comparing the benefits in variance reduction to the cost of dropping principal components.

3.4. Other Related Methods

There are several other rerandomization methods that balance covariates based on some quadratic form. For example, Zhao and Ding (2024) analyze covariate balance using p -values from various statistical tests. Their “joint acceptance rules” based on regressing the treatment vector onto the covariates can be written as a quadratic form using the Wald test statistic $\hat{\boldsymbol{\beta}}^T \hat{\mathbf{V}}^{-1} \hat{\boldsymbol{\beta}}$ where $\hat{\boldsymbol{\beta}}$ is the vector of estimated coefficients and $\hat{\mathbf{V}}$ is the Huber-White robust covariance matrix. In Liu et al. (2025) the authors suggest a prior-induced distance given by $d_\pi = \hat{\boldsymbol{\tau}}_X^T (\boldsymbol{\mu}_\pi \boldsymbol{\mu}_\pi^T + \Sigma_\pi) \hat{\boldsymbol{\tau}}_X$ for some prior distribution with mean $\boldsymbol{\mu}_\pi$ and covariance Σ_π . Finally, Lu et al. (2023) make use of weighted Euclidean distances for rerandomization, which they note are widely used in practice (Li et al. 2017; Hayes and Moulton 2017). These, too, can be expressed as quadratic forms.

4. Quadratic Form Rerandomization

The rerandomization methods in Section 3 all balance some kind of quadratic form, that is, $Q_A(\sqrt{n}\hat{\boldsymbol{\tau}}_X) := (\sqrt{n}\hat{\boldsymbol{\tau}}_X)^T \mathbf{A} (\sqrt{n}\hat{\boldsymbol{\tau}}_X)$ for some positive semi-definite $\mathbf{A} \in \mathbb{R}^{d \times d}$. Therefore, the difference between each of these methods depends on the choice of $\mathbf{A}$, which we summarize in Table 1. Derivations for each table entry are provided in the supplementary material.

Table 1. Choices of $\mathbf{A}$ for various rerandomization methods.

Method	$\mathbf{A}$	Reference
Mahalanobis	Σ^{-1}	Morgan and Rubin (2012, 2015)
Ridge	$(\Sigma + \lambda \mathbf{I}_d)^{-1}$	Branson and Shao (2021)
PCA	$\propto \mathbf{V} \begin{pmatrix} \mathbf{D}_k^{-2} & 0 \\ 0 & 0 \end{pmatrix} \mathbf{V}^T$	Zhang, Yin, and Rubin (2024)
p -value (joint test)	$\propto \Sigma^{-1} \hat{\mathbf{V}}^{-1} \Sigma^{-1}$	Zhao and Ding (2024)
Weighted Euclidean	$\text{diag}\{a_1, \dots, a_d\}$	Lu et al. (2023)

In this section, we establish formal results for rerandomization using any quadratic form $Q_A(\sqrt{n}\hat{\boldsymbol{\tau}}_X)$ as a balance metric where $\mathbf{A} \in \mathbb{R}^{d \times d}$ is a fixed, nonzero, positive semi-definite matrix. This allows us to quickly derive properties of rerandomization schemes not previously considered and determine which choice of $\mathbf{A}$ is optimal for balancing covariates and minimizing the variance of $\hat{\boldsymbol{\tau}}$. We follow the traditional rerandomization procedure: generate potential randomizations until $Q_A(\sqrt{n}\hat{\boldsymbol{\tau}}_X) \leq a$. We choose a based on a given acceptance probability α , such that $\mathbb{P}(Q_A(\sqrt{n}\hat{\boldsymbol{\tau}}_X) \leq a \mid \mathbf{X}) = \alpha$. One can quickly determine a by Monte Carlo simulation; since asymptotically $Q_A(\sqrt{n}\hat{\boldsymbol{\tau}}_X) \mid \mathbf{X} \sim \sum_{j=1}^d \eta_j \mathcal{Z}_j^2$ where $\mathcal{Z}_1, \dots, \mathcal{Z}_d \stackrel{iid}{\sim} \mathcal{N}(0, 1)$ and $\eta_1, \dots, \eta_d$ are the eigenvalues of $\Sigma^{1/2} \mathbf{A} \Sigma^{1/2}$ (Mathai and Provost 1992), we can simulate from this distribution many times and then define a as an empirical quantile of these draws. Alternatively, we can approximate the distribution of $Q_A(\sqrt{n}\hat{\boldsymbol{\tau}}_X) \mid \mathbf{X}$ using an extension of the Welch-Satterthwaite method (Stewart et al. 2007), which we discuss in the supplementary material.

4.1. Theoretical Properties

First, we note that under Quadratic Form Rerandomization all covariate mean differences, whether they are observed or unobserved, are centered at zero, and $\hat{\boldsymbol{\tau}}$ is unbiased due to $Q_A(\sqrt{n}\hat{\boldsymbol{\tau}}_X)$ being symmetric in the treatment assignment $\mathbf{W}$, that is, the acceptance rule $\varphi(\cdot, \mathbf{W})$ satisfies $\varphi(\cdot, \mathbf{W}) = \varphi(\cdot, \mathbf{I}_n - \mathbf{W})$ (Morgan and Rubin 2012). Clearly, treatment symmetry is satisfied since $Q_A(\sqrt{n}\hat{\boldsymbol{\tau}}_X) = Q_A(-\sqrt{n}\hat{\boldsymbol{\tau}}_X)$. Throughout the article we leverage results from Li, Ding, and Rubin (2018) that establish the conditions required for us to characterize asymptotic distributions under Quadratic Form Rerandomization; we describe these conditions in detail in the supplementary material. In short, two conditions are required: Condition 1 states that the finite-population variance and covariance of potential outcomes, individual treatment effects, and covariates are well-defined and have limiting values asymptotically. Meanwhile, Condition 2 ensures that the rerandomization criterion is symmetric in the treatment assignment and places a nonzero probability on some treatment assignments. The following result establishes the covariance of the covariate mean differences under Quadratic Form Rerandomization.

Theorem 1. Let $Q_A(\sqrt{n}\hat{\boldsymbol{\tau}}_X) = (\sqrt{n}\hat{\boldsymbol{\tau}}_X)^T \mathbf{A} (\sqrt{n}\hat{\boldsymbol{\tau}}_X)$ where $\mathbf{A} \in \mathbf{S}_+^d$ and $a > 0$. Then, under Conditions 1 and 2 in the supplementary material, as $n \rightarrow \infty$,

$$\begin{aligned} \text{cov}(\sqrt{n}\hat{\boldsymbol{\tau}}_X \mid \mathbf{X}, Q_A(\sqrt{n}\hat{\boldsymbol{\tau}}_X) \leq a) \\ = \Sigma^{1/2} \Omega \left(\text{diag}\{(\nu_{j,\eta})_{1 \leq j \leq d}\} \right) \Omega^T \Sigma^{1/2} \end{aligned}$$

where $\Omega \in \mathbb{R}^{d \times d}$ is the orthogonal matrix of eigenvectors of $\Sigma^{1/2} \mathbf{A} \Sigma^{1/2}$ and

$$v_{j,\eta} = \mathbb{E} \left[\mathcal{Z}_j^2 \mid \sum_{\ell=1}^d \eta_\ell \mathcal{Z}_\ell^2 \leq a \right] \leq 1 \quad (4)$$

where $\mathcal{Z}_1, \dots, \mathcal{Z}_d \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$ and $\eta_1 \geq \dots \geq \eta_d \geq 0$ are the eigenvalues of $\Sigma^{1/2} \mathbf{A} \Sigma^{1/2}$.

Because the reduction terms $v_{1,\eta}, \dots, v_{d,\eta}$ are not analytically tractable, throughout the article we make use of an approximation for $v_{1,\eta}, \dots, v_{d,\eta}$ derived in the supplementary material of Lu et al. (2023). They show that for $\Sigma^{1/2} \mathbf{A} \Sigma^{1/2} \succ 0$,

$$\begin{aligned} v_{j,\eta} &= \mathbb{E} \left[\mathcal{Z}_j^2 \mid \sum_{\ell=1}^d \eta_\ell \mathcal{Z}_\ell^2 \leq a \right] \\ &= \frac{p_d}{\eta_j} \det(\Sigma^{1/2} \mathbf{A} \Sigma^{1/2})^{1/d} \alpha^{2/d} + o(\alpha^{2/d}) \end{aligned} \quad (5)$$

where $p_d = \frac{2\pi}{d+2} \left(\frac{2\pi^{d/2}}{d\Gamma(d/2)} \right)^{-2/d}$ and $\eta_1, \dots, \eta_d$ are the eigenvalues of $\Sigma^{1/2} \mathbf{A} \Sigma^{1/2}$. Note that the remainder term $o(\alpha^{2/d})$ is taken for a fixed d as $\alpha \rightarrow 0$. Thus, the remainder will be small for α close to zero. The terms p_d and $\alpha^{2/d}$ act as scaling factors depending on the number of covariates and acceptance probability; $v_{j,\eta}$ is increasing in d and α . Intuitively, this result tells us that the choice of $\mathbf{A}$ changes the values of $v_{1,\eta}, \dots, v_{d,\eta}$ primarily through the inverse of its eigenvalues, $\frac{1}{\eta_1}, \dots, \frac{1}{\eta_d}$ and the determinant of $\Sigma^{1/2} \mathbf{A} \Sigma^{1/2}$.

It is useful to reflect on the geometry implied by [Theorem 1](#) in order to better understand the mechanism by which Quadratic Form Rerandomization operates. By [Theorem 1](#), Quadratic Form Rerandomization simply rotates $\Sigma^{1/2}$ alongside the eigenvectors of $\Sigma^{1/2} \mathbf{A} \Sigma^{1/2}$ and scales them by $v_{1,\eta}, \dots, v_{d,\eta}$ (before rotating back). Geometrically, this represents the set of all ellipsoidal constraints on the covariance of the covariate mean differences. While there are other possible shape-based constraints that could be applied (ones that cannot be written as a quadratic form), the ellipsoid is a natural choice as it directly manipulates the eigenstructure of Σ . In Morgan and Rubin (2012), the authors suggest that a benefit of Mahalanobis Rerandomization is that the covariance structure stays the same after rerandomization. Soon, we discuss the costs and benefits of changing the shape of the covariance matrix after rerandomization.

4.2. Choosing the Optimal Quadratic Form for Covariate Balance

Because Quadratic Form Rerandomization depends on the choice of the matrix $\mathbf{A}$, it is natural to wonder what choice is most preferable given the covariates $\mathbf{X}$. Here, we determine which choice of $\mathbf{A}$ is optimal, in the sense of constraining the covariance matrix of the covariate mean differences defined in [Theorem 1](#). Importantly, we explore optimality without assuming that there is any information about the potential outcomes available before the experiment is conducted. Others, such as Lu et al. (2023) and Liu et al. (2025), have derived rerandomization methods that are optimal for variance reduction (under

certain restrictions), but implementing these methods requires information about how the covariates are related to potential outcomes.

We define two ways of quantifying covariance reduction. First, we consider minimizing a measure of the size of the covariance matrix defined in [Theorem 1](#). A natural norm to consider is the Frobenius norm, given by $\|\mathbf{M}\|_F^2 = \sum_{j=1}^d \sigma_j^2(\mathbf{M})$ for some $\mathbf{M} \in \mathbb{R}^{d \times d}$ where $\sigma_j^2(\mathbf{M})$ are the eigenvalues of $\mathbf{M}^T \mathbf{M}$. This is a natural norm to consider because $\mathbf{M}^T \mathbf{M}$ is proportional to the sample covariance matrix of $\mathbf{M}$. Second, we consider maximizing the total variance reduction $\sum_{j=1}^d (1 - v_{j,\eta})$. We will find that $\mathbf{A} = \mathbf{I}_d$ (Euclidean Rerandomization) is optimal for the former, and $\mathbf{A} = \Sigma^{-1}$ (Mahalanobis Rerandomization) is optimal for the latter. In [Section 4.3](#) we consider how reductions to the covariance matrix impact the variance reduction of the difference-in-means estimator. There, we find that $\mathbf{A} = \mathbf{I}_d$ is minimax optimal, in the sense that the difference-in-means estimator's precision is never too far from the optimal choice, regardless of the relationship between covariates and outcomes. The following theorem establishes that the choice $\mathbf{A} = \mathbf{I}_d$ is optimal for minimizing the Frobenius norm of the covariance matrix.

Theorem 2. Let $Q_{\mathbf{I}_d}(\sqrt{n}\widehat{\boldsymbol{\tau}}_{\mathbf{X}}) = \|\sqrt{n}\widehat{\boldsymbol{\tau}}_{\mathbf{X}}\|_2^2$. Then, for all $\mathbf{A} \in \mathbf{S}_+^d$, as $\alpha \rightarrow 0$ and $n \rightarrow \infty$,

$$\begin{aligned} &\|\text{cov}(\sqrt{n}\widehat{\boldsymbol{\tau}}_{\mathbf{X}} \mid \mathbf{X}, Q_{\mathbf{I}_d}(\sqrt{n}\widehat{\boldsymbol{\tau}}_{\mathbf{X}}) \leq a)\|_F \\ &\leq \|\text{cov}(\sqrt{n}\widehat{\boldsymbol{\tau}}_{\mathbf{X}} \mid \mathbf{X}, Q_{\mathbf{A}}(\sqrt{n}\widehat{\boldsymbol{\tau}}_{\mathbf{X}}) \leq a')\|_F + o(\alpha^{2/d}) \end{aligned}$$

under Conditions 1 and 2 in the supplementary material, where a and a' are chosen to have a common acceptance probability α .

Note that while the Frobenius norm grows with the number of covariates, the remainder term will remain small for sufficiently small α . Indeed, in [Section 5](#) we validate via simulation that the inequality in [Theorem 2](#) without the remainder term holds across different covariate dimensions d . To our knowledge, Euclidean Rerandomization has not been previously studied in the literature, so we will take a moment to discuss this method. First, note that our definition of $Q_{\mathbf{I}_d}(\sqrt{n}\widehat{\boldsymbol{\tau}}_{\mathbf{X}})$ requires the covariates to be standardized such that they have unit variance. While other methods such as Mahalanobis Rerandomization are affine invariant, this is not the case for Euclidean Rerandomization—thus, if the covariates have not been standardized, then Euclidean Rerandomization's performance may suffer. Second, because $Q_{\mathbf{I}_d}(\sqrt{n}\widehat{\boldsymbol{\tau}}_{\mathbf{X}}) \mid \mathbf{X} \sim \sum_{j=1}^d \lambda_j \mathcal{Z}_j^2$ where $\lambda_1, \dots, \lambda_d$ are the eigenvalues of Σ , the variance reduction factors under Euclidean Rerandomization are $v_{j,\lambda} = \mathbb{E}[\mathcal{Z}_j^2 \mid \sum_{\ell=1}^d \lambda_\ell \mathcal{Z}_\ell^2 \leq a]$. This manifests in a spherical acceptance region where all directions of variation have equal magnitude after Euclidean Rerandomization. As a result, there is a greater variance reduction applied to the top eigenvectors of Σ , and less applied to the bottom eigenvectors. [Theorem 2](#) implies that in order to control the “size” of the covariance matrix after rerandomization, the best thing we can do is ensure all directions of variation are of equal magnitude.

Whereas Euclidean Rerandomization minimizes the Frobenius norm of the covariance matrix after Quadratic Form Rerandomization, the following theorem establishes that $\mathbf{A} = \Sigma^{-1}$

maximizes the total variance reduction applied to the eigenvectors of $\Sigma^{1/2}\mathbf{A}\Sigma^{1/2}$.

Theorem 3. For all $\mathbf{A} \in \mathbf{S}_+^d$, under Conditions 1 and 2 in the supplementary material, as $\alpha \rightarrow 0$ and $n \rightarrow \infty$, $\sum_{j=1}^d v_a \leq \sum_{j=1}^d v_{j,\eta} + o(\alpha^{2/d})$ where $v_a = \frac{\mathbb{P}(\chi_{d+2}^2 \leq a)}{\mathbb{P}(\chi_d^2 \leq a)}$.

Taken together, [Theorems 2 and 3](#) illustrate that Euclidean Rerandomization and Mahalanobis Rerandomization represent two extremes of a spectrum of rerandomization methods. On one side of the spectrum, Mahalanobis Rerandomization leaves the shape of the covariance matrix unchanged; it only scales the eigenvectors by v_a such that there is an equal variance reduction for all eigenvectors. Meanwhile, Euclidean Rerandomization scales each eigenvector by a different factor $v_{j,\eta}$ such that each of the eigenvectors after rerandomization has the same magnitude.

Ultimately, the method that best reduces the variance of $\hat{\tau}$ depends on the relationship between covariates and outcomes. In the next section, we derive the $\mathbf{A}$ that minimizes the variance of $\hat{\tau}$ for a given relationship between covariates and potential outcomes. We then establish that Euclidean Rerandomization is minimax optimal, in the sense that the variance of $\hat{\tau}$ after Euclidean Rerandomization is never too far from the minimum variance of $\hat{\tau}$, regardless of the covariates' relationship to the potential outcomes.

4.3. Variance Reduction of the Difference-in-Means Estimator $\hat{\tau}$

In the previous section, we established that certain choices of $\mathbf{A}$ are optimal in terms of covariate balance. However, our goal is not just to obtain covariate balance, but also to estimate the average treatment effect (ATE) τ precisely. In this section, we establish how the relationship between covariates and potential outcomes suggests a choice of $\mathbf{A}$ that minimizes the variance of the difference-in-means estimator. To make this connection, we quantify how the covariates are related to the potential outcomes.

Proposition 1. For all $\mathbf{A} \in \mathbf{S}_+^d$, under Conditions 1 and 2 in the supplementary material, and as $n \rightarrow \infty$,

$$\begin{aligned} \text{var}(\sqrt{n}(\hat{\tau} - \tau) \mid \mathbf{X}, Q_{\mathbf{A}}(\sqrt{n}\hat{\tau}_{\mathbf{X}}) \leq a) \\ = \boldsymbol{\beta}^T \text{cov}(\sqrt{n}\hat{\tau}_{\mathbf{X}} \mid \mathbf{X}, Q_{\mathbf{A}}(\sqrt{n}\hat{\tau}_{\mathbf{X}}) \leq a) \boldsymbol{\beta} + \text{var}(\varepsilon) \end{aligned}$$

where $\boldsymbol{\beta}$ is the coefficient vector from the linear projection of $\sqrt{n}(\hat{\tau} - \tau)$ onto $\sqrt{n}\hat{\tau}_{\mathbf{X}}$ under complete randomization, $\varepsilon \sim \mathcal{N}(0, V_{\tau\tau}(1 - R^2))$, and $V_{\tau\tau} = \text{var}(\sqrt{n}(\hat{\tau} - \tau) \mid \mathbf{X})$.

Here, ε encompasses a residual term that is independent of and unaffected by rerandomization. The terms $\boldsymbol{\beta}$, R^2 , and $V_{\tau\tau}$ have been used in other works establishing asymptotic properties of rerandomization (e.g., Li, Ding, and Rubin 2018; Lu et al. 2023). In the supplementary material, we give a precise definition of these terms and a fuller discussion of asymptotic results for rerandomization. Note that [Proposition 1](#) does not assume that a linear model holds. Instead, because Quadratic Form Rerandomization only balances covariate means, the variance reduction is only impacted by the linear relationship between

covariates and outcomes. The following theorem establishes that Quadratic Form Rerandomization always weakly reduces the variance of $\hat{\tau}$ compared to complete randomization. When $\boldsymbol{\beta} \neq \mathbf{0}$, this reduction is strict except in degenerate cases where the rerandomization criterion does not constrain the covariate directions that are predictive of the outcome.

Theorem 4. Suppose $\mathbf{A} \in \mathbf{S}_+^d$. Then, under Conditions 1 and 2 in the supplementary material, the difference $\text{var}(\sqrt{n}(\hat{\tau} - \tau) \mid \mathbf{X}) - \text{var}(\sqrt{n}(\hat{\tau} - \tau) \mid \mathbf{X}, Q_{\mathbf{A}}(\sqrt{n}\hat{\tau}_{\mathbf{X}}) \leq a)$ as $n \rightarrow \infty$ is given by

$$\sum_{j=1}^d ((\Omega^T \mathbf{V})_j \Lambda^{1/2} \boldsymbol{\beta}_Z)^2 (1 - v_{j,\eta}) \geq 0 \quad (6)$$

where $\boldsymbol{\beta}_Z = \mathbf{V}^T \boldsymbol{\beta}$ and $\mathbf{V}$ is the matrix of singular vectors of $\mathbf{X}$, and $(\Omega^T \mathbf{V})_j$ denotes the j th row of $\Omega^T \mathbf{V}$. If Σ and $\Sigma^{1/2}\mathbf{A}\Sigma^{1/2}$ share an eigenbasis, then (6) simplifies to $\sum_{j=1}^d \beta_{Z,j}^2 \lambda_j (1 - v_{j,\eta})$ where $\lambda_1, \dots, \lambda_d$ are the eigenvalues of Σ .

This result clarifies that the variance reduction of $\hat{\tau}$ depends on how each principal component is related to the outcomes, given by $\boldsymbol{\beta}_Z$, as well as the eigenvalues λ_j and variance reduction factors $v_{j,\eta}$. Thus, rerandomization methods that place more importance on reducing the variance of the top k principal components (e.g., Euclidean, Ridge, and PCA Rerandomization) will result in small $\text{var}(\sqrt{n}(\hat{\tau} - \tau) \mid \mathbf{X}, Q_{\mathbf{A}}(\sqrt{n}\hat{\tau}_{\mathbf{X}}) \leq a)$ only if these top components are indeed strongly related to the potential outcomes. On the other hand, if the bottom principal components are relatively important predictors of the potential outcomes, rerandomization methods that do not penalize these terms (such as Mahalanobis Rerandomization) perform better than other methods. We explore this phenomenon further via simulation in [Section 5](#). [Theorem 4](#) quantifies how much Quadratic Form Rerandomization reduces the variance of $\hat{\tau}$, compared to complete randomization. Meanwhile, the following theorem establishes the choice of $\mathbf{A}$ that maximizes this reduction.

Theorem 5. For all $\mathbf{A} \in \mathbf{S}_+^d$ and $\boldsymbol{\beta} \neq \mathbf{0}$, under Conditions 1 and 2 in the supplementary material and a common acceptance probability α , as $n \rightarrow \infty$,

$$\begin{aligned} \text{var}(\sqrt{n}(\hat{\tau} - \tau) \mid \mathbf{X}, Q_{\mathbf{A}^*}(\sqrt{n}\hat{\tau}_{\mathbf{X}}) \leq a) \\ \leq \text{var}(\sqrt{n}(\hat{\tau} - \tau) \mid \mathbf{X}, Q_{\mathbf{A}}(\sqrt{n}\hat{\tau}_{\mathbf{X}}) \leq a') \end{aligned}$$

where $\mathbf{A}^* = \boldsymbol{\beta} \boldsymbol{\beta}^T$. Furthermore, the percent reduction in the variance of $\hat{\tau}$ relative to complete randomization is $100(1 - v_{1,\beta}^* R^2)$ where $v_{1,\beta}^* = \mathbb{E}[\chi_1^2 \mid (\boldsymbol{\beta}^T \Sigma \boldsymbol{\beta}) \chi_1^2 < a]$.

Unlike our previous optimality results, [Theorem 5](#) holds without any kind of remainder. Intuitively, we can see that $\mathbf{A}^*$ directly weights the covariate mean differences by how closely they are related to the potential outcomes. Moreover, $v_{1,\beta}^*$ establishes a direct connection between the variance reduction applied to $\hat{\tau}$ and the eigenstructure of Σ via the weighted eigenvalues $\beta_{Z,1}^2 \lambda_1, \dots, \beta_{Z,d}^2 \lambda_d$. [Theorem 5](#) is similar to [Theorem 4](#) of Lu et al. (2023); however, Lu et al. (2023) require that the covariates are orthogonal and $\mathbf{A}$ is diagonal, whereas we only

require that covariates are standardized and that $\mathbf{A}$ is positive semi-definite. Our result lines up exactly with Theorems 1 and 2 of Liu et al. (2025), who derived the optimal rerandomization strategy when using a Bayesian prior to inform which covariates to balance, among all rerandomization schemes with asymptotic acceptance probability α . However, our Theorem 5 is derived by optimizing over the class of quadratic forms, whereas Liu et al. (2025) optimize over a general class of rerandomization schemes and find that a quadratic form is optimal.

The optimal matrix $\mathbf{A}^*$ in Theorem 5 can only be computed if β is known before the experiment is conducted, and thus typically selecting $\mathbf{A}^*$ is infeasible. However, there are cases where outcome information may be available before the start of a rerandomized experiment, for example, if baseline outcome measures are recorded, or if a smaller pilot experiment was conducted beforehand. In order to provide guidance on how to choose $\mathbf{A}$ when β is unknown, it is useful to quantify the difference between the variance reduction of Quadratic Form Rerandomization using the optimal $\mathbf{A}^*$ and using any other $\mathbf{A}$. The following theorem shows that among all positive semi-definite choices of $\mathbf{A}$ when β is unknown, Euclidean Rerandomization is minimax optimal.

Theorem 6. Suppose $\mathbf{A} \in \mathbf{S}_+^d$, $\beta \neq \mathbf{0}$, and that Conditions 1 and 2 in the supplementary material hold. Then, define

$$\Delta_\beta(\mathbf{A}) = \text{var}(\sqrt{n}(\hat{\tau} - \tau) \mid \mathbf{X}, Q_{\mathbf{A}^*}(\sqrt{n}\hat{\tau}_X) \leq a) \\ - \text{var}(\sqrt{n}(\hat{\tau} - \tau) \mid \mathbf{X}, Q_{\mathbf{A}}(\sqrt{n}\hat{\tau}_X) \leq a')$$

to be the difference in variances under the oracle versus any other quadratic form, assuming a common acceptance probability α . Then, as $n \rightarrow \infty$ and $\alpha \rightarrow 0$,

$$\arg \min_{\mathbf{A} \in \mathbf{S}_+^d} \sup_{\|\beta\|_2 < c} |\Delta_\beta(\mathbf{A})| = \{\mathbf{A} : \Sigma^{1/2} \mathbf{A} \Sigma^{1/2} \propto \Sigma\}.$$

Note that $\mathbf{A} = \mathbf{I}_d$ satisfies the proportionality requirement for minimax optimality. We assume $\|\beta\|_2 \leq c$ because by (6) if β is unbounded the difference in variances may be infinite. Thus, Theorem 6 suggests that Euclidean Rerandomization is a “safe” choice, in the sense that the variance of the difference-in-means estimator is never too far from the optimal choice. We can provide some intuition behind this result by considering the variance reduction factors under Euclidean Rerandomization, $v_{j,\lambda}$, and under the optimal $\mathbf{A}^*$, which we define as $v_{1,\beta}^*$:

$$v_{j,\lambda} = \mathbb{E} \left[\mathcal{Z}_j^2 \mid \sum_{\ell=1}^d \lambda_\ell \mathcal{Z}_\ell^2 \leq a \right] \text{ and} \\ v_{1,\beta}^* = \mathbb{E} \left[\mathcal{Z}_1^2 \mid (\beta^T \Sigma \beta) \mathcal{Z}_1^2 \leq a \right]$$

where $\mathcal{Z}_1, \dots, \mathcal{Z}_d \sim \mathcal{N}(0, 1)$. Informally, these results suggest that if $\beta_{Z,1}^2, \dots, \beta_{Z,d}^2$ are viewed as unknown weights applied to the eigenvalues of Σ , Euclidean Rerandomization ensures that each weight is only off from the optimal choice by at most $\beta_{Z,j}^2$. Since $\mathbf{I}_d$ is positive-definite, these variance reduction factors are distributed across all d principal components, as opposed to $v_{1,\beta}^*$, which is able to concentrate the variance reduction onto a single component.

4.4. Quadratic Form Rerandomization on Principal Components

All of the previous optimality results used all d covariates in the balance metric $Q_{\mathbf{A}}(\sqrt{n}\hat{\tau}_X)$; however, Zhang, Yin, and Rubin (2024) showed that there can be benefits to dropping unimportant covariates during rerandomization. In this section, we consider implementing Quadratic Form Rerandomization on the top k principal components, such that the remaining $d - k$ components receive zero weight. Thus, we extend the PCA Rerandomization methodology of Zhang, Yin, and Rubin (2024) to quadratic forms; we show that this is analogous to considering positive semi-definite $\mathbf{A}$ of rank k . Similar to the definitions given in Section 3.3, let $Q_{\mathbf{A}}^k(\sqrt{n}\hat{\tau}_Z) := \sqrt{n}(\bar{\mathbf{Z}}_T^{(k)} - \bar{\mathbf{Z}}_C^{(k)})^T \mathbf{A} \sqrt{n}(\bar{\mathbf{Z}}_T^{(k)} - \bar{\mathbf{Z}}_C^{(k)})$ be the quadratic form associated with the top k principal components, where $k \leq d$. The following theorem establishes the covariance matrix of the covariate mean differences under Quadratic Form Rerandomization on the principal components and under a positive semi-definite $\mathbf{A} \in \mathbb{R}^{k \times k}$.

Theorem 7. Suppose $\mathbf{A} \in \mathbf{S}_+^k$. Then, under Conditions 1 and 2 in the supplementary material, as $n \rightarrow \infty$,

$$\text{cov}(\sqrt{n}\hat{\tau}_X \mid \mathbf{X}, Q_{\mathbf{A}}^k(\sqrt{n}\hat{\tau}_Z) \leq a) \\ = \Sigma^{1/2} \mathbf{V} \begin{pmatrix} \Omega_k \left(\text{diag}(v_{j,\eta}(k))_{1 \leq j \leq k} \right) \Omega_k^T & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_{d-k} \end{pmatrix} \mathbf{V}^T \Sigma^{1/2}$$

where $v_{j,\eta}(k) = \mathbb{E} \left[\mathcal{Z}_j^2 \mid \sum_{\ell=1}^k \eta_\ell \mathcal{Z}_\ell^2 \leq a \right]$ with $\mathcal{Z}_1, \dots, \mathcal{Z}_k \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$ such that $\eta_1, \dots, \eta_k$ are the eigenvalues of $\Lambda_k^{1/2} \mathbf{A} \Lambda_k^{1/2}$, Ω_k is the matrix of eigenvectors of $\Lambda_k^{1/2} \mathbf{A} \Lambda_k^{1/2}$, Λ_k is the diagonal matrix of the top k eigenvalues of Σ , and $\mathbf{V}$ is the matrix of singular vectors.

If we compare choosing a positive semi-definite $\mathbf{A} \in \mathbb{R}^{d \times d}$ using Theorem 1 with rerandomization using the top k principal components, as defined in Theorem 7, we can see that the covariance structures are analogous; both produce variance reductions $v_{1,\eta}(k), \dots, v_{d,\eta}(k)$, where here k is either the rank of $\mathbf{A}$ or the number of principal components included, but where $v_{j,\eta}(k) = 1$ for $j > k$. This is because when $\mathbf{A}$ has rank $k < d$, then $\eta_j = 0$ for all $j > k$, and thus $v_{j,\eta}(k) = 1$. Importantly, the variance reduction factors $v_{1,\eta}(k), \dots, v_{k,\eta}(k)$ from Theorem 7 are smaller than the reduction factors $v_{1,\eta}, \dots, v_{k,\eta}$ in Theorem 1 under a full d -dimensional quadratic form; this introduces a tradeoff—practitioners gain more reduction to the top k eigenvectors of Σ at the cost of applying no reduction to the bottom $d - k$ eigenvectors.

A natural question raised by Theorem 7 is how one should choose the number of principal components k . Although Zhang, Yin, and Rubin (2024) discuss decision rules for k , their rules are only based on the percentage of variance explained. Instead, we consider selecting principal components via a cost-benefit calculation: we drop a principal component only if the benefit in added variance reduction outweighs the cost of omitting this principal component. The following proposition establishes necessary and sufficient conditions under which the reduced quadratic form has a smaller variance than the full quadratic form.

Proposition 2. Suppose $\mathbf{A}_k \in \mathbf{S}_+^k$, $\mathbf{A}_d \in \mathbf{S}_+^d$, Σ and $\Sigma^{1/2} \mathbf{A}_d \Sigma^{1/2}$ share an eigenbasis, and Λ_k and $\Lambda_k^{1/2} \mathbf{A}_k \Lambda_k^{1/2}$ share an eigenbasis. Then, under Conditions 1 and 2 in the supplementary material and a common acceptance probability α , it follows that $\text{var}(\sqrt{n}(\hat{\tau} - \tau) \mid \mathbf{X}, Q_A^d(\sqrt{n}\hat{\tau}_X) \leq a_d) \geq \text{var}(\sqrt{n}(\hat{\tau} - \tau) \mid \mathbf{X}, Q_A^k(\sqrt{n}\hat{\tau}_Z) \leq a_k)$ as $n \rightarrow \infty$ if and only if

$$\sum_{j=1}^k \beta_{Z,j}^2 \lambda_j (v_{j,\eta}(d) - v_{j,\eta}(k)) \geq \sum_{j=k+1}^d \beta_{Z,j}^2 \lambda_j (1 - v_{j,\eta}(d)).$$

As we show in the supplementary material, the eigenbasis assumption can be removed, in which case the inequality will be notationally similar to (6) in Theorem 4. We invoke the eigenbasis assumption to simplify notation for our decision rules here. Proposition 2 shows that it is preferable to drop the bottom $d - k$ principal components if $\beta_{Z,j}^2 \lambda_j$ is very small for $j \in \{k+1, \dots, d\}$. When β is known, we propose the following decision rule,

$$k = \arg \max_j \left\{ \sum_{i=1}^j \beta_{Z,i}^2 \lambda_i (v_{i,\eta}(d) - v_{i,\eta}(j)) - \sum_{i=j+1}^d \beta_{Z,i}^2 \lambda_i (1 - v_{i,\eta}(d)) \right\} \quad (7)$$

Meanwhile, for unknown β , an analogous decision rule is

$$k = \arg \max_j \left\{ \sum_{i=1}^j \lambda_i (v_{i,\eta}(d) - v_{i,\eta}(j)) - \sum_{i=j+1}^d \lambda_i (1 - v_{i,\eta}(d)) \right\}. \quad (8)$$

Henceforth, we refer to (8) as the Weighted Eigenvalue rule. (7) suggests that, when β is known, it can be preferable in terms of precision to implement Quadratic Form Rerandomization using only the first k principal components, rather than the full set of components d . However, β is typically not known at the start of the experiment. In this case, (8) selects k such that the variance reduction benefits applied to the top principal components maximally outweigh the costs incurred to the bottom principal components. This may result in less precision for the difference-in-means estimator if the bottom principal components are strongly related to the potential outcomes. If information about β is known before the start of an experiment, for example, from a pilot study (Liu et al. 2025), then (7) could be applied using estimates of β .

Equation (8) requires searching all $j = 1, \dots, d$ possible values, which can be computationally intensive. As discussed earlier, an alternative way to upweight the top principal components is via Euclidean Rerandomization, that is, setting $\mathbf{A} = \mathbf{I}_d$. In fact, one could place further preference on the top principal components by setting $\mathbf{A} = \Sigma$, because in this case the eigenvalues of $\Sigma^{1/2} \mathbf{A} \Sigma^{1/2}$ will be $\eta_j = \lambda_j^2$ for $j = 1, \dots, d$. This method is fast to implement, and in Section 5 we find that it performs particularly well in high-dimensional settings, although it may not be as robust as Euclidean Rerandomization. This method can be generalized to any power of λ_j by choosing $\mathbf{A} = \Sigma^c$ for

some $c \geq 0$. In later sections, we refer to this choice of $\mathbf{A}$ as Squared Euclidean Rerandomization when $c = 1$.

4.5. Conducting Inference Post-Rerandomization

The previous sections discuss the design stage of a randomized experiment with Quadratic Form Rerandomization. Here, we discuss the analysis stage—that is, how one conducts inference for the ATE after the experiment has been conducted. There are two primary ways inference can be conducted. The first is with a randomization-based confidence interval (Rosenbaum 2002, chap. 2), which allows us to obtain valid, finite-population inference at the cost of computation time. Randomization-based confidence intervals are often constructed by inverting a sharp null hypothesis that specifies the relationship between potential outcomes, such as $H_0^\tau : Y_i(1) = Y_i(0) + \tau$ for all $i = 1, \dots, n$. Under H_0^τ , we can compute the potential outcomes for any hypothetical randomization and set of observed outcomes. We can conduct a randomization test for H_0^τ as follows:

1. Generate M hypothetical randomizations, where each $\mathbf{w}^{(m)}$ for $m = 1, \dots, M$ is chosen by rerandomizing $\mathbf{W}$ until $Q_A(\sqrt{n}\hat{\tau}_X) \leq a$ for some prespecified $a > 0$.
2. Compute a test-statistic $t(\mathbf{w}, \mathbf{X}, \mathbf{y})$ across all M rerandomizations assuming that H_0^τ is true where $\mathbf{y}$ is the vector of observed outcomes.
3. Compute the randomization-based p -value, defined as $p = \frac{1 + \sum_{m=1}^M \mathbb{1}(|t(\mathbf{w}^{(m)}, \mathbf{X}, \mathbf{y})| > |t^{\text{obs}}|)}{M+1}$,

where t^{obs} is the observed test-statistic (Phipson and Smyth 2010). Then, the randomization-based confidence interval is given by the set of τ such that we fail to reject H_0^τ .

The second way of conducting inference is by using the asymptotic distribution of $\hat{\tau}$, as shown in the supplementary material. In short, one can find that under Conditions 1 and 2, as $n \rightarrow \infty$, $\sqrt{n}(\hat{\tau} - \tau) \mid \mathbf{X}, Q_A(\sqrt{n}\hat{\tau}_X) \leq a \sim \varepsilon + \beta^T \xi \mid \mathbf{X}, Q_A(\xi) \leq a$ (Li, Ding, and Rubin 2018). Here, $\xi \sim \mathcal{N}(\mathbf{0}, \Sigma)$ and $\varepsilon \sim \mathcal{N}(0, V_{\tau\tau}(1 - R^2))$ where $V_{\tau\tau} = \frac{1}{p} S_{Y(1)}^2 + \frac{1}{1-p} S_{Y(0)}^2 - S_\tau^2$ such that $S_{Y(\cdot)}^2$ are the finite population variances of the outcomes in treatment and control and S_τ^2 is the finite-population variance of the individual treatment effects. In practice, $S_{Y(1)}^2$, $S_{Y(0)}^2$, and R^2 are estimated with sample analogs. However, note that $S_\tau^2 = \frac{1}{n-1} \sum_{i=1}^n (\tau_i - \tau)^2$ is not estimable, so estimated intervals will be conservative. From here, we can easily obtain confidence intervals for τ by computing the quantiles of $\sqrt{n}(\hat{\tau} - \tau) \mid \mathbf{X}, Q_A(\sqrt{n}\hat{\tau}_X) \leq a$, which can be done via Monte Carlo simulation.

5. Simulations and Application

To better understand how different rerandomization methods compare in practice in terms of covariate balance and treatment effect estimation, we consider a simulation study and real data application that compares several Quadratic Form Rerandomization methods: Euclidean, Mahalanobis, PCA (using the Mahalanobis distance on the top k principal components with k chosen by both the Kaiser rule and the weighted eigenvalue rule in (8)), Squared Euclidean Rerandomization, and the oracle $\mathbf{A}^* = \beta\beta^T$.

5.1. Design of the Simulated Data

As discussed in Section 4, the eigenstructure of Σ plays an important role in determining the performance of rerandomization methods. To better understand how the eigenstructure impacts the performance of rerandomization methods, we simulate the eigenvalues of Σ from a symmetric Dirichlet distribution with concentration parameter γ , scaled by the number of covariates d : that is, $\lambda_1, \dots, \lambda_d \sim d \cdot \text{Dirichlet}(\gamma)$. As a result, $\sum_{j=1}^d \lambda_j = d = \text{tr}(\Sigma)$. As $\gamma \rightarrow 0$, the eigenvalues become sparser and as $\gamma \rightarrow \infty$, the eigenvalues are more uniform; we consider concentration values $\gamma \in \{0.05, 0.5, 1\}$. In what follows, we refer to $\gamma = 0.05$ as “sparse” concentration, $\gamma = 0.5$ as “average”, and $\gamma = 1$ as “uniform.”

We can use this assumed distribution on the eigenvalues to gain intuition about the variance reduction factors $v_{1,\eta}, \dots, v_{d,\eta}$ for a given $\mathbf{A}$. When the eigenvalues of $\Sigma^{1/2} \mathbf{A} \Sigma^{1/2}$ are not all equal, there is a differential percentage variance reduction applied to the covariates. As an illustration when $d = 50$, Figure 1 shows that methods such as Euclidean and Squared Euclidean Rerandomization provide a much greater variance reduction to the top eigenvectors, but much less to the bottom eigenvectors. Meanwhile, PCA Rerandomization applies two levels of variance reduction—one level to the top k principal components, and no reduction to the bottom $d - k$ principal components. Mahalanobis Rerandomization applies a constant level of variance reduction across all eigenvalues.

After simulating the eigenvalues $\lambda_1, \dots, \lambda_d$ for a given d , we simulate our covariates as $\mathbf{X} \sim \mathcal{N}(\mathbf{0}, \Sigma)$ where $\Sigma = \mathbf{P}(\text{diag}\{\lambda_1, \dots, \lambda_d\})\mathbf{P}^T$ and $\mathbf{P} \in \mathbb{R}^{d \times d}$ is a random orthogonal matrix. After simulating $\mathbf{X}$, we center and standardize each column such that each has mean zero and variance one. For each simulation, we generate $n = 500$ observations where 250 units are assigned to treatment and 250 units are assigned to control. We define the potential outcomes $Y_i(0)$ and $Y_i(1)$ for unit i as $Y_i(0) \sim \mathcal{N}(\mathbf{Z}\boldsymbol{\beta}_Z, 1)$ and $Y_i(1) = Y_i(0) + \tau$ where $\tau = 1$, $\boldsymbol{\beta}_Z \in \mathbb{R}^d$, and $\mathbf{Z} = \mathbf{X}\mathbf{V}$ is the matrix of principal components of $\mathbf{X}$. For each choice of $\gamma \in \{0.05, 0.5, 1\}$ and $d \in \{5, 25, 50, 75, 100, 150, 200, 250\}$ we simulate 10,000 datasets and implement Mahalanobis, Euclidean, PCA, Squared Euclidean, and the oracle rerandomization for each. We set $\alpha = 0.01$ and consider three choices of $\boldsymbol{\beta}_Z$:

- (i) $\boldsymbol{\beta}_Z = (\lambda_d, \dots, \lambda_1)^T$, which preferentially weights the bottom principal components.

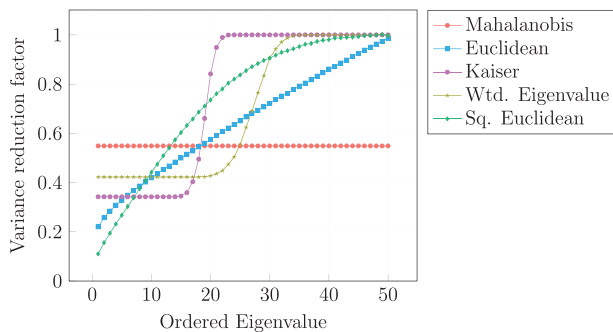


Figure 1. $v_{j,\eta}$ averaged over 10,000 draws of $\lambda_1, \dots, \lambda_d \sim d \cdot \text{Dirichlet}(\gamma = 1)$.

- (ii) $\boldsymbol{\beta}_Z = (1, \dots, 1)^T$, which equally prioritizes each principal component.
- (iii) $\boldsymbol{\beta}_Z = (\lambda_1^{1/2}, \dots, \lambda_d^{1/2})^T$, which preferentially weights the top principal components.

5.2. Results: Variance Reduction of $\hat{\tau}$

Figure 2 displays the ratio between the standard deviation of $\hat{\tau}$ (y-axis) under rerandomization versus complete randomization for (i), (ii), and (iii) (rows) across 10,000 simulated datasets, for different numbers of covariates (x-axis) and concentration γ (columns).

We can see that Mahalanobis Rerandomization is the best (besides the oracle) in terms of variance reduction in (i), whereas Euclidean Rerandomization is the best at constraining the variance in (ii). In setting (iii), the Mahalanobis distance performs the worst, in the sense that it yields the largest relative standard error, whereas Euclidean Rerandomization is still a close second in terms of variance reduction, reflecting the robustness of Euclidean Rerandomization. PCA Rerandomization using the decision rule in (8) performs better than the Kaiser rule when eigenvalues are sparse, but the two methods are approximately equivalent when the eigenvalues become more uniform. Furthermore, as the eigenvalues become more uniform there is less information that can be derived from the covariance matrix, and as a result all rerandomization methods are closer to each other. Finally, we note that the oracle is relatively more precise as d increases, which follows due to the increasing R^2 and decreasing $v_{1,\beta}^*$ across d , as illustrated by Theorem 5.

Importantly, we can see that practitioners should take great care when the distribution of eigenvalues is sparse. In this case, there are large differences among rerandomization methods; Mahalanobis Rerandomization does better than all other methods in setting (i), but significantly worse in settings (ii) and (iii). Euclidean Rerandomization is a good hedge against this risk, as it constrains the covariance matrix the most, and produces the second greatest amount of variance reduction across all settings. This aligns with Theorem 6 which showed that Euclidean Rerandomization is minimax optimal in terms of the variance of the difference-in-means estimator $\hat{\tau}$. Furthermore, it is notable that Squared Euclidean Rerandomization and Euclidean Rerandomization are nearly identical in performance across simulations. Thus, methods that upweight top eigenvalues like Squared Euclidean Rerandomization may outperform Euclidean Rerandomization when the top principal components are more strongly related to the potential outcomes, but may underperform if the bottom principal components have significant importance.

5.3. Application

So far, our simulations have only considered the setting where the covariates are linearly related to the potential outcomes and the treatment effect is additive. Here, we relax those assumptions using a real experiment and constructing synthetic outcomes. In our real data example, we consider a randomized controlled trial in which 308 low-socioeconomic-status German adolescents were randomly assigned mentors to help them

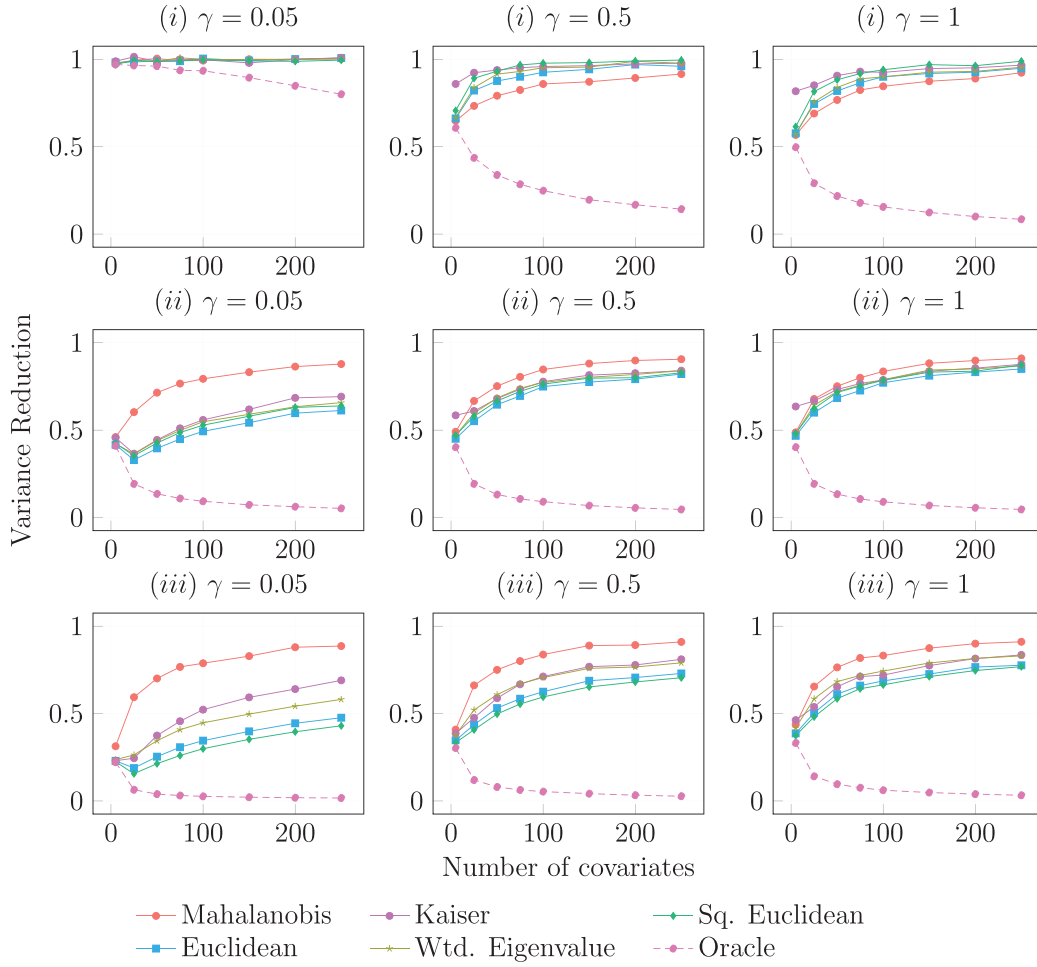


Figure 2. Variance reduction across different numbers of covariates and concentrations.

prepare for professional life, where randomization followed a pairwise matching design with rerandomization to ensure covariate balance (Resnjanskij et al. 2024). We include $d = 21$ covariates—these include continuous variables such as age and personality scores, and categorical variables such as grade level, sex, or migrant status. We recode all categorical variables with binary indicator variables for each category. Furthermore, if any categorical variables were missing, we denoted their missingness with indicators. The experiment yielded outcomes Y_i , a continuous measure of labor market outcomes. In reality, we only know the outcomes Y_i and not the full potential outcomes $Y_i(1), Y_i(0)$. For the purposes of comparing rerandomization methods, we generate potential outcomes in a way that induces non-linearity and non-additivity. For each subject i , we define the potential outcomes $Y_i(0) = \mathbb{E}[Y_i | X_i, W_i = 0]$ and $Y_i(1) = \mathbb{E}[Y_i | X_i, W_i = 1]$ where we use flexible, non-linear models for estimating the regression function via the SuperLearner package in R (Polley et al. 2025; van der Laan, Polley, and Hubbard 2007). Specifically, we used an ensemble modeling approach with a candidate library that included random forests, generalized additive models, gradient boosted models, LASSO, and generalized linear models. This yields data $(X_i, W_i, Y_i(0), Y_i(1))$ such that the data are non-linearly related to the potential outcomes and the treatment effect is not additive.

Table 2. Precision of $\hat{\tau}$ under rerandomization relative to complete randomization.

Method	$\text{var}(\hat{\tau} X, Q_A(\sqrt{n}\hat{\tau}_X) \leq a) / \text{var}(\hat{\tau} X)$
Mahalanobis	0.611
Euclidean	0.565
Kaiser ($k = 8$)	0.585
Weighted Eigenvalue ($k = 16$)	0.608
Sq. Euclidean	0.576

Once covariates and outcomes are defined, we ran each Quadratic Form Rerandomization method 10,000 times with acceptance probability $\alpha = 0.01$, and computed the difference-in-means estimator $\hat{\tau}$ for each replication. For comparison, we did the same for 10,000 complete randomizations. Table 2 shows how the precision of the difference-in-means estimator compares across rerandomization methods, relative to complete randomization. Table 2 shows how the precision of the difference-in-means estimator compares across rerandomization methods, relative to complete randomization, where the bolded value highlights the method with the highest relative precision.

Although performance was close across methods, Euclidean Rerandomization yielded the smallest variance. The reason for the close comparison between methods likely stems from the eigenstructure of the covariates, as illustrated by the scree plot in Figure 3. Since the distribution of eigenvalues is very uniform, there is less information from the covariance matrix that can be

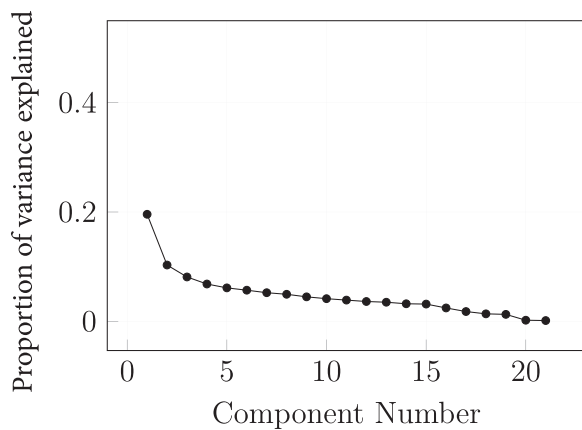


Figure 3. Scree plot for the $d = 21$ standardized covariates in the mentoring data set. The x-axis indexes the principal components in decreasing order of their eigenvalues. The y-axis reports the proportion of total covariate variance explained by each component.

leveraged during rerandomization. Informally, this is a setting close to the one described in Section 5.2 when $\gamma = 1$. Regardless, the results in Table 2 align with Theorem 6, which shows that Euclidean Rerandomization is minimax optimal and thus, its variance reduction is more robust than other methods across many settings.

6. Discussion and Conclusion

Much of the rerandomization literature has focused on applying the Mahalanobis distance to different experimental settings (Morgan and Rubin 2012, 2015; Branson, Dasgupta, and Rubin 2016; Li, Ding, and Rubin 2018; Zhou et al. 2018; Lu et al. 2023). However, the Mahalanobis distance can perform poorly in high-dimensional settings. Recent works such as Branson and Shao (2021) and Zhang, Yin, and Rubin (2024) have attempted to address this problem by using quadratic forms other than the Mahalanobis distance as a balance metric for rerandomization. In our analysis, we derive general results for rerandomization using any quadratic form and establish which method is optimal under different conditions.

Under Quadratic Form Rerandomization, defined by a positive semi-definite matrix $\mathbf{A}$, we show that the variance reduction applied to each principal component of $\mathbf{X}$ is weighted by the eigenvalues of $\Sigma^{1/2}\mathbf{A}\Sigma^{1/2}$ where $\Sigma = \text{cov}(\sqrt{n}(\bar{\mathbf{X}}_T - \bar{\mathbf{X}}_C) \mid \mathbf{X})$. In terms of covariate balance, we found in Section 4.2 that $\mathbf{A} = \mathbf{I}_d$ minimizes the Frobenius norm of the covariance of the covariate mean differences after rerandomization, and $\mathbf{A} = \Sigma^{-1}$ maximizes the total variance reduction applied to the principal components. Meanwhile, in Section 4.3, we showed that the precision of the difference-in-means estimator after rerandomization depends on the variance explained by each principal component and their relationship with the potential outcomes. We found that Euclidean Rerandomization is minimax optimal in the sense that the difference-in-means estimator's precision is never too far from its precision under the optimal choice of $\mathbf{A}$. Our results establish properties about a general class of rerandomization methods and provide guidance for navigating this class in practice.

We validated these theoretical results via simulation and a real application, and found that—while the optimal rerandomization method depends on the unknown relationship between the principal components and outcomes—Euclidean Rerandomization yielded fairly robust results across simulation settings. That said, if the practitioner has prior information that the top principal components are strongly related to the outcomes, they can employ methods like Squared Euclidean or PCA Rerandomization to achieve greater variance reduction than Euclidean Rerandomization, but at the expense of robustness.

Although this work establishes properties for a broad class of rerandomization methods, we only focus on methods that can be expressed by quadratic forms. Future work could establish methods for comparing rerandomization with quadratic forms to other rerandomization methods and experimental design strategies. Furthermore, although we focus on two-arm, completely randomized experiments, the quadratic form framework is not tied to this setting. For example, in multi-armed experiments one could stack imbalances across arms and apply quadratic form acceptance rules to this higher-dimensional covariate imbalance vector. Furthermore, Quadratic Form Rerandomization can be layered on top of other randomization schemes (e.g., stratified/block, cluster, or split-plot) by extending the approaches in Wang, Wang, and Liu (2023), Lu et al. (2023), and Shi, Zhao, and Liu (2025); these are all natural directions for future work. In particular, sequential rerandomization could be especially useful, as practitioners may be able to use outcome information in prior experiments to inform future experiments. This outcome information could be useful not only for choosing which covariates to balance, but also which quadratic form to use when rerandomizing.

Supplementary Materials

The supplementary materials include: regularity conditions for asymptotic results, a discussion of the geometry and eigenstructure of the covariance of the covariate mean differences under Quadratic Form Rerandomization, a discussion of the distribution of quadratic forms and how to simulate acceptance thresholds in practice, and proofs for all results in the main text.

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